Tanmay Bhuskute

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SUMMARY

Infrastructure Engineer and M.S. Computer Science candidate (Dec 2026) with production experience automating and provisioning cloud infrastructure using Python, Go, Terraform, and Ansible across AWS environments. Skilled in Kubernetes orchestration, CI/CD pipelines, and building internal platforms for engineering and cross-functional teams.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Go, Java, JavaScript, TypeScript, C/C++, SQL.
Infrastructure & Cloud: AWS, Google Cloud Platform, Azure, Terraform, Ansible, Docker, Kubernetes, ArgoCD, Helm.
Developer Tools: Git, GitLab CI/CD, Jenkins, Linux, Bash, HashiCorp Vault, SonarQube.
Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs.
Databases: PostgreSQL, MySQL.

WORK EXPERIENCE

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure provisioning workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** and enabling reliable rollback.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault; built self-service developer platform adopted by **15 engineers**.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage LLM automation pipeline orchestrating device-level test execution for Android bug reproduction, evaluated across 53 runs on 20+ apps against a published GPT-4o baseline.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI, Docker] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.
- Containerized and deployed serving API with Docker on Oracle Cloud, orchestrating data pipeline, model, and REST layer as independent services.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP, Docker] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).